Sri Chaitanya IIT Academy., India.

A right Choice for the Real Aspirant
Regional Office: Chennai - Tamilnadu

2D- STRAIGHT LINES & CIRCLES
DAY -01 (15-03-24)

SECTION – I

(Multiple Correct Answers Type)

1. If the straight line joining origin to the points of inter section of the line $\frac{x}{a}+\frac{y}{b}=1$ and the circle $5\left(x^2+y^2+ax+by\right)=8ab$ are at right angles then $\frac{a}{b}=$

 A)1 B)-1 C)$\frac{1}{2}$ D)2

2. A straightline L passing through the point $A(-2,-3)$ cuts the lines $x+3y=9$ and $x+y+1=0$ at B and C respectively If $(AB)(AC)=20$ Then equation of L is

 A)$x-y=1$ B)$x-y+1=0$ C)$3x-y+3=0$ D)$3x-y=3$

3. The equation of the circle touching the line $2x-3y+1=0$ at (1,1) and having radius $\sqrt{13}$

 A)$x^2+y^2-6x+4y=0$ B)$x^2+y^2=13$
 C)$x^2+y^2+2x-8y+4=0$ D)$x^2+y^2+2x+10y-13=0$

4. Let P be a point inside the equilateral $\triangle^{le} ABC$ such that PA=3 , PB=4 , PC=5 then

 A)The side of the equilateral triangle ABC is $\sqrt{25+12\sqrt{3}}$

 B)The radius of in circle of triangle ABC is $\sqrt{3}\left(\frac{\sqrt{25+12\sqrt{3}}}{2}\right)$

 C)The radius of nine point circle of triangle ABC is $\left(\frac{\sqrt{25+12\sqrt{3}}}{2\sqrt{3}}\right)$

 D)Area of triangle ABC is $\frac{\sqrt{3}}{4}\left(25+12\sqrt{3}\right)$

5. If D(3,4), E(5,7), F(1,5) divides the sides BC,CA,AB respectively in the same ratio 3:1 then

 A)area of triangle ABC is 7/4 sq.units B)Centroid of triangle ABC is $\left(3,\frac{16}{3}\right)$

C)Triangle DEF is isosceles D)area of triangle DEF (4) sq.units

6. $P(2,3)$ is a given point and $a(x+2y-3)+b(x+y-2)=0$ is a family of lines. If L_1, L_2 of the family respectively are at maximum and minimum distance from P then

A) Equation of L_1 is x + 2y – 3 = 0

B) Equation of L_2 is 2x – y – 1 = 0

C) The distance of orthocentre of Δ formed by L_1, L_2 and x – axis from origin is $\sqrt{2}$

D) The circumcentre of triangle formed by L_1, L_2 and x – axis is $\left(\frac{7}{4}, 0\right)$

7. If the equation $x^2+2\sqrt{2}xy+2y^2+4x+4\sqrt{2}y+1=0$ represents the tangents to same circle then

A) Diameter of the circle is 1 unit

B) Area of the circle is π sq. units

C) Equation of one line passing through centre of the circle is $x+\sqrt{2}y+2=0$

D) Perimeter of the circle is π sq. units

8. $A(1,2)$ and $B(7,10)$ are two fixed points. If $P(x,y)$ is such that a point where $\underline{|APB} = 60^0$ and the area of ΔAPB is maximum then the point P

A) is on the line $3x+4y=36$

B) is on any line perpendicular to AB

C) is on perpendicular bisector of AB

D) is on the circle passing through $(1,2)$ and $(7,10)$ having radius 10

9. C_1, C_2 are two circles of radii $a, b\ (a<b)$ touching both coordinates axes and have their centers in first quadrant, then which of the following statement is/are true?

A) If C_1 and C_2 touch each other then $\frac{b}{a}=3+2\sqrt{2}$

B) If C_1 and C_2 are orthogonal then $\frac{b}{a}=2+\sqrt{3}$

C) If C_1 and C_2 intersect the such a way that their common chord has maximum length then $\frac{b}{a}=3$

D) If C_2 passes through centre of C_1 then $\frac{b}{a}$ can be $2+\sqrt{2}$

10. Two distinct chords each bisected by x – axis are drawn to the circle $2\left(x^2+y^2\right)-2ax-by=0$ from the point $\left(a, b/2\right)$. Then which of the following statements hold good?

A) $\frac{a^2}{b^2}=3$ B) $\frac{a^2}{b^2}=4$ C) $\frac{a^2}{b^2}=\frac{5}{2}$ D) $\frac{a^2}{b^2}=\frac{1}{2}$

11. From an arbitrary point P on the circle $x^2+y^2=4$, tangents are drawn to the circle $x^2+y^2=1$, which meet the former circle again at points A & B. The locus of the point of intersection of tangents drawn to the former circle at A & B is

A) A line B) a circle of radius 3 unit

C) An ellipse D) a conic with center at the origin

12. The circle $x^2+y^2=4$, cuts the line joining the points A(1, 0) & B(3, 4) in two points P & Q. Let $\frac{BP}{PA}=\alpha>0$ & $\frac{BQ}{QA}=\beta>0$. Then,

A) $\alpha+\beta=5$ B) $\alpha+\beta=\frac{16}{3}$ C) $\alpha\beta=21$ D) $\alpha\beta=7$

13. Co-ordinates of a point at an unit distance from the lines $3x-4y+1=0$ and $8x+6y+1=0$, are

A) $\left(\frac{6}{5},\frac{-1}{10}\right)$ B) $\left(\frac{-8}{5},\frac{3}{10}\right)$ C) $\left(0,\frac{3}{2}\right)$ D) $\left(\frac{-2}{5},\frac{-13}{10}\right)$

14. In ΔABC, $A\equiv(\alpha, \beta)$, $B\equiv(1, 2)$, $C\equiv(2, 3)$ and the point A lies on the line $y=2x+3$, where α, β are integers, area of the triangle is S, such that $[S]=2$, where $[.]$ denotes the greatest integer function. Which of these can be possible coordinates of A ?

A) $(-7, -11)$ B) $(-6, -9)$ C) $(2, 7)$ D) $(3, 9)$

15. Let ABCD be a square of side length 4units. C_1 is the circle through vertices A, B, C, D and C_2 is the circle touching all the sides of the square ABCD. If P is a point on C_2 and Q is the another point on C_1, then $\dfrac{PA^2+PB^2+PC^2+PD^2}{QA^2+QB^2+QC^2+QD^2}$ can be

A) 0.75 B) 1.25 C) 1 D) 0.5

16. $P(2,3)$ is a given point and $a(x+2y-3)+b(x+y-2)=0$ $(a,b\in R)$ is family of lines. If L_1, L_2 of the family respectively are at maximum and minimum distance from P then

A) Equation of L_1 is $x+2y-3=0$

B) Equation of L_2 is $2x-y-1=0$

C) The distance of orthocentre of Δ^{le} formed by L_1, L_2 and x-axis from origin is $\sqrt{2}$

D) The circumcentre of triangle formed by L_1, L_2 and x-axis is $\left(\frac{7}{4},0\right)$

17. Circle(s) touching the x-axis at a distance of 3 units from the origin and having an intercept of length $2\sqrt{7}$ on the y-axis is (are)

A) $x^2+y^2-6x+8y+9=0$ B) $x^2+y^2-6x+7y+9=0$

C) $x^2+y^2-6x-8y+9=0$ D) $x^2+y^2-6x-7y+9=0$

18. Consider the straight line $x+2y+4=0$ and $4x+2y-1=0$. The line $6x+6y+7=0$ is

A) Bisector of the angle containing origin

B) Bisector of acute angle

C) Bisector of obtuse angle

D) Bisector of the angle not containing origin

19. Which of the following is/are false?

A) A parallelogram circumscribing a circle is a rhombus

B) Angle between two tangents drawn from an external point to a circle is complementary to the angle subtended by the line segment joining the points of contact at the centre

C) Through a point T, a tangent T A and a secant TPQ are drawn to the circle passing through the points A, Q & P. If the chord AB is drawn parallel to PQ, then triangles PAT and BAQ are congruent

D) The equation of circumcircle of a ΔABC is $x^2+y^2+3x+y-6=0$. If A=(1,-2), B=(-3,2) And the vertex C varies then the locus of orthocenter of ΔABC is a circle.

SECTION – II

(Integer Answer Type)

1. If two circles of unit radius intersect orthogonally and the common area is $\frac{\pi}{\lambda}-1$ *then* $\lambda=$

2. Area bounded by the minimum $\{|x|,|y|\}=1$ *and the* $\min\{|x|,|y|\}=2$ *is*

3. If P and Q are variable points on $C_1: x^2+y^2=4$ *and* $C_2: x^2+y^2-8x-6y+24=0$ respectively then maximum value of PQ is equal to

4. A circle C of radius unity lies in first quadrant and touches both the co-ordinate axes. If the radius of the circle which touches both co-ordinate axes and passes through centre of C has radius either r_1 or r_2 $(r_1 \neq r_2)$ then $\frac{|r_1-r_2|}{\sqrt{2}}$ equals

5. In ΔABC, the co-ordinates of the vertex A are (4,-1) and lines $x-y-1=0$ and $2x-y=3$ are internal bisectors of angles B and C. Then radius of in-circle of ΔABC is $\frac{k}{\sqrt{5}}$ where k=

6. From a variable point P(t, t) ; $t \in R$ & |t|>1, a pair of tangents are drawn to the circle $x^2+y^2=1$, touching it at points A&B then, the locus of the centroid of the ΔPAB is a part of the line whose slope is___

7. Six distinct points (x_i, y_i), i = 1, 2,...,6 are taken on the circle $x^2+y^2=4$ such that $\sum_{i=1}^{6} x_i = 8$ & $\sum_{i=1}^{6} y_i = 4$. The line joining orthocenter of a triangle formed by any three points out of these six and the centroid of the triangle having the vertices at the rest of these points, passes through a fixed point (h, k) then h+k is________

8. If the area of a hexagon, whose vertices taken in order be (5, 0), (4, 2), (1, 3), (-2, 2), (-3, -1) and (0, -4), is A sq unit, then the digit at unit's place of A is

9. A variable straight line of slope 4, intersects the curve xy=1 at two points A & B. If the locus of the point which divides the line segment between A & B in the ratio 1:2 is $16x^2+10xy+y^2=k$, then k = _____

10. One diagonal of a square is the portion of $\frac{x}{97}+\frac{y}{79}=1$ intercepted between the axes. P_1, P_2 are the lengths of the perpendiculars from the vertices present on the other diagonal on the axis of Y, then $P_1^2+P_2^2$ is equal to λ, then the digit at the unit place of λ is

11. Consider a Circle $x^2+y^2-4x-6y+11=0$ if $m_1 \& m_2$ are the maximum and minimum values of $\frac{y}{x}$ for all ordered pairs (x,y) on the circumference of the circle, then the value of m_1+m_2 is

12. Tangents are drawn to the circle $x^2+y^2=1$ at the points where it is met by the circles $x^2+y^2-(\lambda+6)x+(8-2\lambda)y-3=0$, λ being a real parameter. The locus of point of intersection of these tangents is $px+qy+10=0$, then $p+q$ is

13. If $A(4,0)$ and $B(9,0)$ and $C(0,h)$ are 3-points such that AB subtends greatest angle at C. If $h>0$, then value of h is

14. In a right angled triangle BC=5, AB=4, AC=3, Let S be the circum circle Let S_1 be the circle touching both sides AB and AC and circle S internally. Let S_2 be the circle touching the sides AB and AC of Δ^{le} ABC, and touching the circle S externally if r_1, r_2 are radii of circles S_1 and S_2 respectively, then $\sqrt{1+r_1r_2}$ is

15. A circle passing through the vertex C of a rectangle and touches its sides AB and AD at M and N respectively. If the $\perp^{lar}$ distance from C to the line segment $\overline{MN}$ is equal to 5 and area of rectangle ABCD is Δ, then $\sqrt{\Delta}$ is

16. Let ABCD be a quadrilateral with area 18 sq,units, with side AB parallel to the side CD, and AB=2CD. Let AD be perpendicular to AB and CD. If a circle is drawn inside the quadrilateral ABCD touching all the sides and its radius is k, then K^2+5 is

17. If the area of the triangle formed by the line $x+y=3$ and the angle bisectors of the pair of lines $x^2-y^2+4y-4=0$ is A, then the value of 16A is

18. ABC be a triangle with $A(1,3)$, $y=x, y=-2x$ are the equations of internal angular bisectors of $\angle B \, and \, \angle C$ and area of triangle ABC is k units, then 2k is

SECTION – III

(Paragraph Type)

PASSAGE-1

P is a variable point on the line L=0. Tangents are drawn to the circle $x^2+y^2=4$ from P and touches the circle at Q and R , the parallelogram PQSR is completed

1. If $L=2x+y-6=0$ then locus of circumcentre of $\Delta^{le}PQR$ *is*

A) $2x-y=4$ B) $2x+y=3$ C) $x-2y=4$ D) $x+2y=3$

2. If $p=(2,3)$ then circumcentre of $\Delta^{le}QRS$ *is*

A) $\left(\frac{2}{13},\frac{7}{26}\right)$ B) $\left(\frac{2}{13},\frac{3}{26}\right)$ C) $\left(\frac{3}{13},\frac{9}{26}\right)$ D) $\left(\frac{3}{13},\frac{2}{13}\right)$

3. If $p=(3,4)$ then coordinates of s are

A) $\left(\frac{-46}{25},\frac{-63}{25}\right)$ B) $\left(\frac{-51}{25},\frac{-68}{25}\right)$ C) $\left(\frac{-46}{25},\frac{-68}{25}\right)$ D) $\left(\frac{-68}{25},\frac{-51}{25}\right)$

PASSAGE-2

The vertex A of ΔABC is $(3,-1)$. Equation of median BE and internal angular bisector CF are $6x+10y-59=0$ and $x-4y+10=0$ respectively. Then……

4. Equation of side AB must be…………

A) $x+y=2$ B) $18x+13y=41$

C) $23x-y-70=0$ D) $x+3y=0$

5. Length of side AC must be……….

A) $\sqrt{83}$ B) $\sqrt{85}$ C) $\sqrt{71}$ D) $\sqrt{62}$

PASSAGE-3

Let the lines represented by the equation $x^2y^2-x^2-y^2+1=0$, form a square ABCD. A point 'P' is taken on the same plane of square such that atleast two sides of all the triangles PAB, PBC, PCD and PDA are equal.

6. The number of possible positions of the point ‘P’

A) 1 B) 5 C) 17 D)9

7. The number of possible positions of point ‘P’ such that atleast one of four given triangles is equilateral is

A) 0 B) 4 C) 8 D) 5

SECTION - IV

(MATRIX MATCHING ANSWER TYPE)

1. Match List – I with List – II and select the correct answer using the code given below lists

	List – I		List – II
(P)	Center of circle inscribed in a square formed by lines $x^2-8x+12=0$ and $y^2-14y+45=0$	(1)	$(4, 0)$
(Q)	A circle passes through $\left(3, \sqrt{\frac{7}{2}}\right)$ and touches $x+y=1,\ x-y=1$. Centre of circle is………	(2)	$(7, 3)$
(R)	A square is inscribed in circle $x^2+y^2-10x-6y+30=0$. One side of square is parallel to $y=x+3$, then one vertex of square could be……..	(3)	$\left(\frac{2}{13}, \frac{3}{13}\right)$
(S)	The circle $x^2+y^2=4$ cuts the circle $x^2+y^2+2x+3y-5=0$ in A and B, then centre of circle with AB as diameter is………	(4)	$(4, 7)$

A) $P-1;Q-3;R-4;S-2$ B) $P-4;Q-2;R-1;S-3$

C) $P-4;Q-1;R-2;S-3$ D) $P-2;Q-1;R-3;S-4$

2. If (x, y) is a point on circle of radius unity and centre $(2, 2)$.

Match the following expressions of List – I with their respective maximum values from List – II

Match List – I with List – II and select the correct answer using the code given below lists

	List – I		List – II
(P)	$x+y$	(1)	$\frac{9+4\sqrt{2}}{2}$
(Q)	$\lvert x-y\rvert$	(2)	$4+\sqrt{2}$
(R)	xy	(3)	$\sqrt{2}$
(S)	$\frac{x}{y}$	(4)	$\frac{4+\sqrt{7}}{3}$

A) $P-2;Q-1;R-3;S-4$ B) $P-2;Q-3;R-1;S-4$

C) $P-1;Q-2;R-3;S-4$ D) $P-4;Q-2;R-1;S-3$

3. ABCD is a rectangle in clockwise direction $A=(1,3)$, $C=(5,1)$ and B, D are on the line $y=2x+C$ then match List – I with List – II and select the correct answer using the code given below lists

	List – I		List – II
(P)	Middle point of BD	(1)	$(2,0)$
(Q)	Middle point of AB	(2)	$(3,2)$
(R)	Middle point BC	(3)	$\left(5/2, 7/2\right)$
(S)	The point D	(4)	$\left(9/2, 5/2\right)$

A) $P-3;Q-2;R-4;S-1$ B) $P-2;Q-3;R-1;S-4$

C) $P-2;Q-3;R-4;S-1$ D) $P-4;Q-3;R-2;S-1$

4. Match List – I with List – II and select the correct answer using the code given below lists

	List – I		List – II
(P)	The circle $x^2+y^2-4x-4y+4=0$ is inscribed in a triangle which has two sides along coordinate axes. The locus of circumcentre of triangle is $x+y-xy+k\sqrt{x^2+y^2}=0$. Then $2k=$	(1)	4

(Q)	A variable circle passes through $A(a, b)$ and touches x – axis. Locus of other end of diameter through A is $(x-a)^2 = \lambda y$ where $\left\|\frac{\lambda}{b}\right\| = \ldots\ldots.$	(2)	2
(R)	If two circles $(x-1)^2 + (y-3)^2 = r^2$ and $x^2 + y^2 - 8x + 2y + 8 = 0$ intersect in two distinct points the number of integer values taken by radius r is………..	(3)	3
(S)	The circle $x^2 + y^2 = 1$ cuts x – axis at P and Q. Another circle with centre at Q and variable radius intersects the first circle at R above x – axis and line segment PQ at S. If the maximum area of ΔQSR is λ then $3\sqrt{3}\,\lambda = \ldots\ldots\ldots$	(4)	5

A) $P-2;Q-1;R-3;S-4$

B) $P-2;Q-1;R-4;S-1$

C) $P-3;Q-1;R-4;S-2$

D) $P-4;Q-2;R-4;S-2$

5.

	Column I		Column II
A)	The area bounded by the curve given by the equation:- $\max\{\|x\|,\|y\|\} = \sqrt{2}$, is equal to _____	P)	1
B)	A straight line cuts the sides BC, CA and AB of a triangle ABC at P,Q & R respectively. If $\|BP\|\|CQ\|\|AR\| = K\|PC\|\|QA\|\|RB\|$, then K=	Q)	3
C)	If the area enclosed by the curve $4\|x-1\| + 3\|y-1\| = 12$, is N sq. unit, then the sum of digits of N is_____	R)	6
D)	If the area of the region bounded by $\|x\| \le 10, \|y\| \le 10$ and $\|x+y\| \le 10$, is M sq. unit, then the digit at hundred's place of M is _____	S)	8

6.

	Column I		Column II
A)	If the equation of the circle passing through the points of intersection of circles $x^2+y^2-4x-6y-12=0$ and $x^2+y^2+6x+4y-12=0$ and cutting $x^2+y^2-2x+3=0$, orthogonally, is $x^2+y^2+\alpha x+7y-12=0$ then α =_____	P)	9
B)	If $4l^2-5m^2+6l+1=0$, then $lx+my+1=0$, touches a fixed circle, whose square of the radius is _____	Q)	5
C)	A circle has the center at (2, 1) and one of the chord is a diameter of the circle $x^2+y^2-2x-6y+6=0$, then the radius of the circle is________	R)	3
D)	If the equation of the locus of mid-points of chords of the circle $x^2+y^2=4$, which substend right angle at the center, is $x^2+y^2=K$, then K = _____	S)	2

7.

	Column I		Column II
(P)	The circle $x^2+y^2-4x-4y+4=0$ is inscribed in a triangle which has two sides along coordinate axes. The locus of circumcentre of triangle is $x+y-xy+k\sqrt{x^2+y^2}=0$. Then 2K = …..	(1)	4
(Q)	A variable circle passes through $A(a,b)$ and touches x-axis. Locus of other end of diameter through A is $(x-a)^2=\lambda y$ where $\left\lvert\frac{\lambda}{b}\right\rvert=..$	(2)	2

(R)	If two circle $(x-1)^2+(y-3)^2=r^2$ and $x^2+y^2-8x+2y+8=0$ intersect in two distinct points the number of integer values taken by radius r is…..	(3)	3
(S)	Area of the region $\lvert 3x-4y\rvert+\lvert 4x+3y\rvert=5\sqrt{2}$ is (sq.units)	(4)	5

A) P-2 ; Q-1 ; R-3 ; S-4 B) P-2 ; Q-1 ; R-4 ; S-1

C) P-3 ; Q – 1 ; R- 4 ; S-2 D) P-4 ; Q-2 ; R-4 ; S-2

8. C_1,C_2 are two circles of radii a,b $(a<b)$ respectively touching both the coordinate axes and have their centres in the first quadrant.

	Column I		Column II
(A)	C_1,C_2 touches each other then $\frac{b}{a}$ is	(P)	$2+\sqrt{3}$
(B)	If C_1,C_2 are orthogonal then $\frac{b}{a}$ is	(Q)	$2+\sqrt{2}$
(C)	If C_1,C_2 intersect in such a way that their common chord has maximum length then $\frac{a}{b}$ is	(R)	3
(D)	If C_2 passes through centre of C_1 then $\frac{b}{a}$ is	(S)	$3+2\sqrt{2}$

A) A-S ; B-R ; C-P ; D-Q B) A-Q ; B-P ; C-R ; D-S

C) A-P ; B-S ; C-Q ; D-R D) A-S ; B-P ; C-R ; D-Q

9.

	Column I		Column II
(A)	The length of the common chord of two circles of radii 3 units and 4 units which intersect orthogonally is $k/5$. Then k is equal to	(P)	1

(B)	The circumference of the circle $x^2+y^2+4x+12y+p=0$ is bisected by the circle $x^2+y^2-2x+8y-q=0$ then $p+q$ is equal to	(Q)	24
(C)	the number of distinct chords of the circle $2x\left(x-\sqrt{2}\right)+y(2y-1)=0$ where the chords are passing through the point $\left(\sqrt{2},1/2\right)$ and are bisected on the x-axis is	(R)	32
(D)	One of the diameters of the circle circumscribing the rectangle ABCD is $4y=x+7$. If A and B are the points $(-3,4)$ and $(5,4)$ respectively, then the area of the rectangle is	(S)	36

A) A-Q,B-S,C-P,D-R
B) A-P,B-S,C-Q,D-R
C) A-Q,B-R,C-P,D-S
D) A-R,B-P,C-Q,D-S

KEY

(MULTIPLE CORRECT CHOICE TYPE)

Q.No.	KEY	Q.No.	KEY	Q.No.	KEY	Q.No.	KEY
1.	CD	6.	ABCD	11.	CD	16.	ABCD
2.	AC	7.	BC	12.	BD	17.	AC
3.	AC	8.	AC	13.	ABCD	18.	AB
4.	AD	9.	ABCD	14.	ABCD	19.	BC
5.	ABD	10.	ABC	15.	A	20.	

(INTEGER ANSWER TYPE)

Q.No.	KEY	Q.No.	KEY	Q.No.	KEY	Q.No.	KEY
1.	2	6.	1	11.	6	16.	9

2.	4	7.	3	12.	1	17.	8
3.	8	8.	4	13.	6	18.	3
4.	2	9.	2	14.	5	19.	
5.	6	10.	5	15.	5	20.	

(Paragraph Type Questions)

Q.No.	KEY	Q.No.	KEY
1.	B	6.	D
2.	C	7.	C
3.	B	8.	
4.	B	9.	
5.	B	10.	

(Matrix Type Questions)

Q.No.	KEY	Q.No.	KEY
1.	C	6.	A-P, B-Q, C-R, D-S
2.	B	7.	B
3.	C	8.	D
4.	B	9.	A
5.	A-S, B-P, C-R, D-Q	10.	